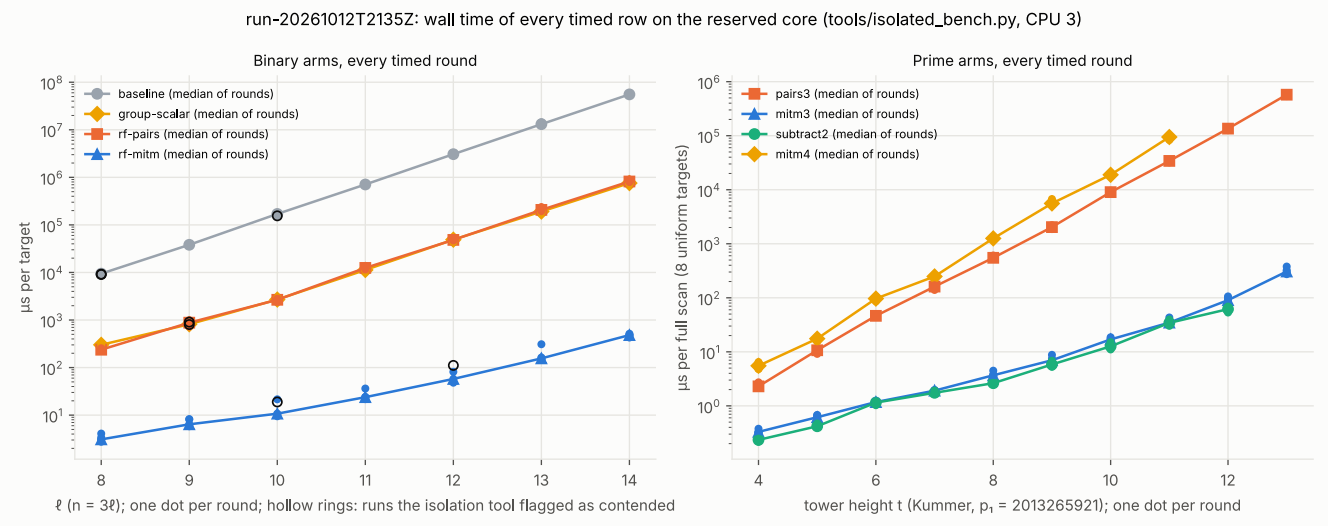
Run report: the isolated timing stage of `pdp_rootfind`, run-20261012T2135Z

Observations only. Interpretation: `../../README.md` and `../../REPORT.pdf`.

What ran

- **Command:** `research/pdp_root_finder_20261012/run_timing.sh 5` from the repository root, started 2026-10-12T21:35:53Z, finished 2026-10-12T22:25:23Z. Every timed command ran through `tools/isolated_bench.py run --cpus 3 --label <label>` (CPU 3 reserved, every other thread moved off it, a 2 s settle check before each run, PSI and other-process CPU recorded; a run whose record says `contended` is listed and not pooled), with `RAYON_NUM_THREADS=1`.
- **Binary:** `target/release/examples/pdp_rootfind` (release build of `examples/pdp_rootfind/` as committed in this PR, base commit `cd0286878`); the `analyze` subcommand of the same source tree is not part of any timed arm.
- **Host:** Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU (no SMT siblings reported), L2 2 MiB a core, L3 260 MiB, 15 GiB RAM, kernel 6.18.44, Python 3.13 (the isolation tool). Isolation level: a reserved core on a shared virtual machine (the tool's records in `../../raw/isolated.jsonl`).
- **Rounds:** five, arms interleaved within a round (A B C D per rung, rung after rung), an A/A pair of the baseline at $\ell = 10$ opening every round.
- **Binary workload** (`bin-time`): arms `baseline` (PS-0′s `decompose`), `group-scalar` (PS-2′s row kernel), `rf-pairs`, `rf-mitm`; rungs $\ell = 8…14$ ($n = 3\ell$) on PS-0′s corpora ($\ell \le 12$) and this study's ($\ell = 13, 14$), and the (23, 8) cell; the first 16 refuted uniform targets of a rung, 3 passes ($\ell \le 11$), 1 pass ($\ell = 12$), 4 targets and 1 pass ($\ell \ge 13$). The unit is ns per target (total wall of all calls ÷ calls); the table arm's table build is recorded apart as `setup_secs` and is not in the per-target figure.
- **Prime workload** (`prime-ladder`): Kummer towers at $p_1 = 2013265921$, seed 20261012, 16 uniform + 8 planted targets a rung; arms `pairs3`, `mitm3` ($m = 3$, $t = 4…13$), `subtract2` ($m = 2$, $t = 4…12$), `mitm4` ($m = 4$, $t = 4…11$); the first 8 uniform targets timed in full-scan mode (every candidate probed, the cost of a refutation), passes repeated until 0.3 s. Unit: ns per full scan.
- **Requested and completed:** 185 commands requested (170 binary `bin-time`, 15 `prime-ladder`); 177 ran, all with exit status 0; 8 were refused before starting by the isolation tool's settle check (other processes used 0.25–0.38 CPU s in the 2 s window, limit 0.20; the top process in every refusal was the session's own agent process `claude`): `baseline-l8` (round 5); `group-scalar-l8` (rounds 2, 4, 5); `rf-pairs-l8` (round 5); `baseline-l9` (round 3); `group-scalar-l9` (round 3); `group-scalar-l12` (round 1). Of the 177 runs the tool flagged 7 as contended (other-process CPU above the limit during the run; the same agent process), listed and not pooled: `round2-aa2-baseline-l10` (0.31 s of other CPU in 2.64 s); `round2-baseline-l8` (0.14 s of other CPU in 0.45 s); `round2-group-scalar-l9` (0.01 s of other CPU in 0.04 s); `round2-rf-mitm-l12` (0.11 s of other CPU in 0.33 s); `round4-baseline-l8` (0.36 s of other CPU in 0.45 s); `round4-rf-pairs-l9` (0.01 s of other CPU in 0.05 s); `round4-rf-mitm-l10` (0.06 s of other CPU in 0.02 s). Every prime-ladder command ran in every round.

Measured values



Every timed row of run-20261012T2135Z: µs per target (binary) and per full scan (prime) on the reserved core, one dot per round, medians joined; hollow rings mark runs the isolation tool flagged as contended.

Binary, ns per refuted target (median and minimum over the uncontended rounds; the counted operations of the same targets beside them):

arm	n	ℓ	ns / target, median	ns / target, min	rounds pooled	contended (excluded)	mul / target	lookups / target
baseline	24	8	9,296,856.0	9,072,994.2	4	0	—	—

group-scalar	24	8	276,453.3	251,695.8	2	0	—	—
rf-mitm	24	8	3,091.5	2,756.1	5	0	637	256
rf-pairs	24	8	235,159.1	224,759.3	4	0	83,069	—
baseline	27	9	37,705,461.6	36,993,764.8	4	0	—	—
group-scalar	27	9	809,708.4	803,516.8	4	0	—	—
rf-mitm	27	9	6,410.9	5,833.6	5	0	1,262	506
rf-pairs	27	9	878,996.3	727,873.5	5	0	322,319	—
baseline	30	10	169,656,215.1	155,094,496.8	15	0	—	—
group-scalar	30	10	2,693,236.3	2,562,275.9	5	0	—	—
rf-mitm	30	10	10,736.5	9,460.7	5	0	2,317	928
rf-pairs	30	10	2,662,909.6	2,445,414.1	5	0	1,080,653	—
baseline	33	11	707,969,176.1	698,012,671.0	5	0	—	—
group-scalar	33	11	11,290,448.6	11,014,276.8	5	0	—	—
rf-mitm	33	11	23,865.1	21,682.6	5	0	5,032	2,014
rf-pairs	33	11	12,462,862.3	11,533,557.6	5	0	5,079,305	—
baseline	36	12	3,069,594,280.9	3,033,374,129.4	5	0	—	—
group-scalar	36	12	48,094,418.5	43,846,688.3	4	0	—	—
rf-mitm	36	12	57,639.9	48,083.3	5	0	9,957	3,984
rf-pairs	36	12	48,374,142.2	46,916,171.2	5	0	19,858,245	—
baseline	39	13	13,115,950,202.5	12,835,260,809.8	5	0	—	—
group-scalar	39	13	191,078,485.0	179,628,647.0	5	0	—	—
rf-mitm	39	13	155,965.0	145,803.2	5	0	20,257	8,104
rf-pairs	39	13	206,909,013.0	194,379,323.5	5	0	82,129,985	—
baseline	42	14	55,506,121,996.2	54,678,643,345.8	5	0	—	—
group-scalar	42	14	757,478,104.0	737,726,535.5	5	0	—	—
rf-mitm	42	14	481,991.0	428,569.2	5	0	40,922	16,370
rf-pairs	42	14	818,311,477.0	793,896,131.8	5	0	335,044,787	—
baseline	23	8	8,698,054.6	8,250,982.8	5	0	—	—
group-scalar	23	8	213,550.8	211,737.6	5	0	—	—
rf-mitm	23	8	3,900.0	2,578.4	5	0	592	238
rf-pairs	23	8	202,207.4	169,462.3	5	0	71,873	—

Prime, ns per full scan (median and minimum over the uncontended rounds):

m	arm	t	ns / full scan, median	ns / full scan, min	rounds pooled	mul / target	table build s
2	subtract2	4	234.6	216.6	5	22	—
2	subtract2	5	419.2	394.0	5	67	—
2	subtract2	6	1,149.2	1,028.5	5	217	—
2	subtract2	7	1,743.7	1,715.6	5	372	—
2	subtract2	8	2,628.4	2,453.7	5	567	—
2	subtract2	9	5,832.1	5,274.7	5	1,237	—
2	subtract2	10	12,635.1	11,071.1	5	2,582	—
2	subtract2	11	34,341.7	30,239.6	5	5,067	—
2	subtract2	12	61,684.7	53,374.8	5	10,392	—
3	mitm3	4	330.0	301.8	5	32	—
3	mitm3	5	616.1	582.3	5	77	—
3	mitm3	6	1,176.0	1,083.6	5	167	0.0001

3	mitm3	7	1,911.9	1,799.0	5	302	0.0003
3	mitm3	8	3,679.9	3,487.1	5	632	0.0009
3	mitm3	9	6,985.7	6,416.5	5	1,207	0.0033
3	mitm3	10	16,968.7	14,933.6	5	2,597	0.0159
3	mitm3	11	34,700.8	33,818.5	5	5,097	0.0680
3	mitm3	12	90,380.0	83,920.9	5	10,147	0.3136
3	mitm3	13	306,503.5	273,533.7	5	20,417	1.7380
3	pairs3	4	2,295.8	2,188.8	5	305	—
3	pairs3	5	10,693.9	9,162.0	5	1,421	—
3	pairs3	6	46,356.7	45,670.7	5	6,083	0.0001
3	pairs3	7	161,260.1	139,568.2	5	19,151	0.0003
3	pairs3	8	549,497.5	544,577.8	5	81,785	0.0009
3	pairs3	9	2,033,676.8	1,869,743.2	5	294,995	0.0033
3	pairs3	10	9,030,183.4	8,409,667.9	5	1,356,677	0.0159
3	pairs3	11	34,147,483.3	32,097,618.2	5	5,211,177	0.0680
3	pairs3	12	135,260,479.9	132,538,994.9	5	20,622,767	0.3136
3	pairs3	13	574,094,285.6	540,701,063.5	5	83,432,033	1.7380
4	mitm4	4	5,518.1	4,815.5	5	495	—
4	mitm4	5	17,517.1	14,814.4	5	1,818	—
4	mitm4	6	96,639.7	71,677.4	5	9,608	0.0001
4	mitm4	7	248,568.0	216,034.5	5	25,358	0.0002
4	mitm4	8	1,257,021.6	1,073,314.1	5	127,800	0.0009
4	mitm4	9	5,574,072.4	5,136,107.5	5	595,193	0.0040
4	mitm4	10	18,900,772.9	17,873,125.0	5	2,147,310	0.0148
4	mitm4	11	94,566,632.1	83,510,969.6	5	9,615,903	0.0781

A/A control (the baseline against itself at $\ell = 10$, every round): 171.0 / 159.1, 156.7 / 155.1, 172.9 / 176.6, 170.1 / 161.1, 174.7 / 169.7 ms per target (rounds 1–5; the round-2 second run is one of the contended runs above); within a pair at most 7.5 % apart, 155–177 ms across the ten runs.

Records

`../../raw/bin_time.jsonl` (one row per timed binary command), `../../raw/prime_time.jsonl` (one row per rung and arm), `../../raw/isolated.jsonl` (the isolation tool's record of every run: label, command, reserved CPUs, settle check, PSI before and after, other processes' CPU seconds, the `contended` flag, exit status), `../../raw/timing_progress.txt`. Graph source: `plot.py` (reads the raw records only). Medians, minima and ratios: `../../ANALYSIS.json` and `../../ANALYSIS.md`, written by `pdp_rootfind analyze`.